

MUONS: THEORY FOR EXPERIMENT

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I. INTRODUCTION AND OBJECTIVES

The primary objective of the research effort described in this proposal is an understanding of the structure of the fundamental physics from experimental data obtained by the muon experiments at the Fermi National Accelerator Laboratory (FNAL). The proposed research is strongly aligned with the 2024 recommendations of the Particle Physics Project Prioritization Panel (P5) and recommendations of the 2021-22 Snowmass study on the future directions for US high energy physics.

An essential component of the success of the program of muon experiments at FNAL, such as Mu2e and its future upgrades, depends on close interactions between experimentalists and theorists. Theoretical computations are needed to separate possible signals of New Physics (NP) from the plethora of non-perturbative QCD effects at several physics scales and in computing background effects that can mimic the signal. While consistent theoretical programs that address the former exist, systematic theoretical studies of experimental backgrounds are often missing. This proposal is designed to close this gap.

Besides theoretical activities at the University of South Carolina (USC) and FNAL, we propose to organize a collaboration of theorists working closely with the Mu2e collaboration on physics and background computations and implementations of those in the Monte Carlo generators deployed by Mu2e. We are planning focused yearly progress meetings at USC or FNAL to discuss the progress with the computations, new problems and opportunities, and implementation.

This project will not involve the construction, alteration, maintenance and/or repair of public infrastructure in the United States.

II. SCIENTIFIC BACKGROUND

Lepton flavor-violating (LFV) processes offer exciting possibilities to probe New Physics (NP) at multi-TeV scales in low-energy experiments. Any NP scenario that involves LFV interactions can be matched to an effective Lagrangian, $\mathcal{L}_{\text{eff}}$, whose Wilson coefficients would be determined by the ultraviolet physics that becomes active at some high scale Λ . This Lagrangian can be conveniently identified by the amount of change of lepton quantum numbers,

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{eff}}^{\Delta L_\mu=0} + \mathcal{L}_{\text{eff}}^{\Delta L_\mu=1} + \mathcal{L}_{\text{eff}}^{\Delta L_\mu=2} = \sum_i \frac{C_i}{\Lambda^2} \mathcal{Q}_i, \quad (1)$$

where $\mathcal{Q}_i$ are the effective operators that are written entirely in terms of the SM degrees of freedom such as leptons: $\ell_i = \tau, \mu$, and e ; and quarks: b, c, s, u , and d . All operators in Eq. (1) are of dimension-6 and have the form $\mathcal{Q} \sim (\bar{\mu}\Gamma e)(\bar{q}\Gamma q)$, where Γ are different Dirac structures. The contributions of the higher-dimensional (or more exotic lepton and baryon-number violating) operators are also possible and can be computed. The C_i are the Wilson coefficients (WC) that are different for different NP models. Experimentally constraining C_i 's probes various NP scenarios, which is the final goal of LFV experiments. Note that the leading term in $\mathcal{L}_{\text{eff}}^{\Delta L_\mu=0}$ is SM, so it is suppressed by M_W , not Λ .

A promising way to approach this goal is to study charged lepton flavor violating (CLFV) transitions using atomic nuclei, as in the muon conversion process $\mu^- + (A, Z) \rightarrow e^- + (A, Z)$ or $\mu^- + (A, Z) \rightarrow e^+ + (A, Z - 2)$, where a muon is changing to an electron without producing neutrinos [1]. The experiment Mu2e, currently under construction at FNAL, will be completed

and will start taking data in 2027. Mu2e is using Al nucleus as a target. An upgrade of this experiment (called Mu2e-II), requires additional R&D effort to improve the constraints on possible NP effects. It will also be using a different target nucleus. The further proposed Advanced Muon Facility (AMF) will exploit Fermilab's future accelerator upgrades and allow studies of CLFV and muonium oscillations at rates far beyond those achievable at any other planned facility.

There are several types of new physics particles that can be probed in LFV experiments. Depending on their masses, they can either contribute to the muon conversion process via Wilson coefficients of effective operators mediating the process or, for the light NP particles, directly.

The first type of NP particles can be taken into account in the framework of the Standard Model Effective Field Theory (SMEFT). Those heavy NP particles are integrated out at the scales associated with their masses, giving contributions to the Wilson coefficients of SMEFT operators. The beauty of this approach is realized in the fact that a finite number of effective operators parameterize all models of NP that can mediate lepton flavor-violating processes [2]. Those operators can be then run down to the low energy scales associated with the muon conversion experiment and matched to the low energy Lagrangian $\mathcal{L}_{\text{eff}}$, written in terms of the degrees of freedom relevant to the muon conversion experiment.

The effective Lagrangian, $\mathcal{L}_{\text{eff}}^{\Delta L_\mu=1}$, can be divided into the dipole part ($\mathcal{L}_D$), and parts describing four-fermion interactions ($\mathcal{L}_{\ell q}$ and $\mathcal{L}_{4\ell}$) as $\mathcal{L}_{\text{eff}}^{\Delta L_\mu=1} = \mathcal{L}_D + \mathcal{L}_{4\ell} + \mathcal{L}_{\ell q}$. The dipole part $\mathcal{L}_D$ is

$$\mathcal{L}_D = -\frac{m_2}{\Lambda^2} \left[\left(C_{DR}^{\ell_1 \ell_2} \bar{\ell}_1 \sigma^{\mu\nu} P_L \ell_2 + C_{DL}^{\ell_1 \ell_2} \bar{\ell}_1 \sigma^{\mu\nu} P_R \ell_2 \right) F_{\mu\nu} + h.c. \right], \quad (2)$$

where $P_{R,L} = (1 \pm \gamma_5)/2$ is the right (left) chiral projection operator. The Wilson coefficients would, in general, be different for different lepton flavors ℓ_i . In turn, $\mathcal{L}_{\ell q}$ can be written as [3],

$$\mathcal{L}_{\ell q} = -\frac{1}{\Lambda^2} \sum_{\Gamma=1, \gamma_5, \gamma_\mu, \dots, \sigma_{\mu\nu}} \sum_{q_1, q_2} \left[\left(C_{VR}^{q_1 q_2 \ell_1 \ell_2} \bar{\ell}_1 \Gamma P_R \ell_2 + C_{VL}^{q_1 q_2 \ell_1 \ell_2} \bar{\ell}_1 \Gamma P_L \ell_2 \right) \bar{q}_1 \Gamma q_2 + h.c. \right]. \quad (3)$$

$\mathcal{L}_{4\ell}$ can be written by substituting quark fields q_i by the lepton ones ℓ_i in Eq. (3). Muon conversion processes employ scattering on atomic nuclei to probe LFV interactions, as in $\mu^- + (A, Z) \rightarrow e^- + (A, Z)$ or $\mu^- + (A, Z) \rightarrow e^+ + (A, Z - 2)$ [4]. Alternatively, the constraints on the Wilson coefficients of the above Lagrangians can be obtained by studying lepton and meson decays [5]. It is interesting to point out that constraints on the underlined physics in $\mu^- + (A, Z) \rightarrow e^+ + (A, Z - 2)$ can, in principle, also be obtained from the process $pp \rightarrow \mu^+ e^+$, which can be studied at Super-Kamiokande. These processes have different theoretical and experimental uncertainties and limitations [6].

The second type of new physics particles that can be probed in the muon conversion experiments include light NP states, i.e. states whose masses are such that they could be produced in these experiments [7]. Mu2e and other future muon conversion experiments at Fermilab can probe the parameter space of such models, providing constraints that are complementary to current astrophysical as well as laboratory searches. The models that can be probed include those with axion-like particles (ALPs) with flavor-violating couplings, massive Z' bosons, and heavy neutral leptons.

While extremely promising in terms of effective luminosity, muon conversion on nuclei also involves physical effects that complicate the proper interpretation of discovery in such experiments. First, many quark-level effective operators contribute to these processes. Second, theoretical predictions for conversion rates in such processes contain uncertainties due to nucleon and

nuclear binding effects. Currently, such effects are estimated with the help of nucleon effective field theories and phenomenological nuclear models [8]. An alternative approach to take into account nuclear binding will be discussed in Sect. IV H.

III. RELEVANT EXPERIENCE AND SUMMARY OF RECENT ACCOMPLISHMENTS RELEVANT TO THIS PROJECT

A proper interpretation of the results of muon conversion experiments is complicated and requires close collaboration of theorists and experimentalists. First, many different sub-processes contribute to these processes, disentangling which requires careful theoretical analysis. This way, one can decipher the structure of possible new interactions that manifested itself in the results of the measurements. Proper identification of those new interactions is the goal of Mu2e, Mu2e-II, and AMF experiments, which require extensive theoretical input. Members of the USC particle and nuclear theory group (A. Petrov) have extensive experience modeling new physics effects both in effective field theories (EFTs) and in the framework of particular NP models (V. Gudkov, M. Schindler). Their expertise will be complemented by other theorists who will be participating in the proposed theory-experiment collaboration that will be described in Sect. IV A. Finally, this theoretical work will be incorporated into the Mu2e analysis framework by the experimental physics member (D.J. Tedeschi).

A. Previously published related work

Phenomenology of Lepton Flavor Violating Interactions: LFV processes offer interesting possibilities to probe NP at scales Λ of several TeVs. A recent surge of interest in LFV processes is supported by a possible connection between lepton flavor universality (LFU) anomalies in decays of heavy flavored mesons and LFV (see, e.g. [9]). If NP effects couple preferentially to the third generation, say via the effective operator in the gauge (primed) basis, $\mathcal{O} \sim G/\Lambda^2 (\bar{b}'\Gamma b')(\bar{\tau}'\Gamma\tau')$, then switching to the mass basis would generally induce both LFV and LFU-violating operators. Because of the tiny SM contributions, observation of charged lepton flavor violating (CLFV) transitions provides especially clean probes of NP. CLFV transitions will be probed in many experiments in the next five years, including Mu2e (FNAL) and COMET (J-PARC, Japan).

Effective field theories encode all possible heavy degrees of freedom that might mediate LFV at low energies as Wilson coefficients of an effective Lagrangian. Studies of low-energy LFV decays of mesons and baryons allow for extracting Wilson coefficients of the operators parameterizing LFV interactions. While this approach is completely independent of the choice of a particular NP model, it makes it difficult to obtain constraints on the individual Wilson coefficients because of the large number of operators contributing to experimental observables. Thus, besides muon conversion $\mu + (A, Z) \rightarrow e + (A, Z)$ these operators can also be probed in other experiments.

For instance, together with graduate student Hazard, Petrov employed the restricted kinematics of exclusive meson decays, which reduces the number of operators to those allowed by the quantum numbers of the decaying states. This can be seen in two-body decays $M \rightarrow \bar{\ell}_1 \ell_2$ of quarkonium states M with different quantum numbers [3]. They also argued that studies of radiative LFV $M \rightarrow \gamma \bar{\ell}_1 \ell_2$ decays could provide important complimentary access to those effective operators [10] and introduced a projection operator formalism to ensure gauge invariance of the decay amplitudes.

Lepton flavor-violating transitions and gluonic operators at Mu2e and tau decays. Together with a former postdoc D. Zhuridov, Petrov discussed the role of gluonic Rayleigh-type operators in effective field theory analysis of LFV transitions at Mu2e experiment [11]. These operators are induced by integrating out heavy quarks that are kinematically inaccessible at the scale where low-energy experiments take place, which makes those experiments sensitive to the couplings of lepton flavor changing neutral currents to heavy quarks. They discussed constraints on the Wilson coefficients of those operators from the muon conversion $\mu + (A, Z) \rightarrow e + (A, Z)$ and from lepton flavor-violating tau decays.

Nonstandard neutrino interactions and transition magnetic moments: Some SMEFT operators involve left-handed lepton doublets, which implies that the same Wilson coefficient governs transitions of both charged leptons and neutrinos. Together with a former graduate student K. Healey and postdoc D. Zhuridov, Petrov considered constraints on generic nonstandard neutrino interactions from existing experimental data on neutrino transition magnetic moments and derived strong bounds on tensorial couplings of neutrinos to charged fermions [12]. They also discussed how some of these tensorial couplings can be constrained by other experiments, e.g., on neutrino-electron and neutrino-nucleus scattering.

Muonium-antimuonium oscillations: An especially clean system to study BSM effects in the lepton sector is muonium M_μ , a QED bound state of a positively-charged muon and a negatively-charged electron. The average lifetime of a muonium state τ_{M_μ} is rather large, $\tau_\mu = (2.1969811 \pm 0.0000022) \times 10^{-6}$ s [4], which allows for precision measurements of muonium properties [13].

One way to search for LFV in muonium is to study muonium-antimuonium oscillations. This process violates the muon lepton number by two units and could be sensitive to the types of NP that are not probed by other types of LFV processes. Conlin and Petrov have computed mass and lifetime differences of muonium states that result from $M_\mu - \overline{M}_\mu$ mixing in effective field theory [14]. The lifetime difference was discussed for the first time. They showed that the leading contribution to the lifetime difference scales as $1/(M_W^2 \Lambda^2)$, so it can be probed in muonium oscillation experiments [15]. New experiments studying muonium are proposed at J-PARC (Japan) and CSNS (China) [16] laboratories. Such experiments can also be performed at Fermilab (AMF) [17].

Electromagnetic interactions of neutrinos: Together with Bansal, Paz, Tammara, and Zupan, Petrov studied NP-induced neutrino-photon couplings. They showed that potentially measurable neutrino polarizability (realized through dimension 7 Rayleigh operators) can be induced in the models with light (pseudo)scalar mediators. They explored the rich phenomenology of such models, computing in detail the constraints on the viable parameter space, and built several explicit models that lead to enhanced neutrino polarizability [18].

Semileptonic decays of heavy mesons and artificial neural networks (ANN): Checks of the second-row unitarity of the Cabibbo-Kobayashi-Maskawa (CKM) matrix involve extractions of the matrix element V_{cd} , which may be obtained from $D \rightarrow \pi$ semileptonic decays. These decay rates are proportional to hadronic form factors that parametrize how the quark $c \rightarrow d$ transition is realized in $D \rightarrow \pi$ meson decays. A question might be asked if choosing a specific functional form for the form factor induces a bias in interpreting the results of experimental analysis. This question was addressed by Petrov and graduate students Grant and Gunawardana in the framework of the machine learning (ML) approach [19], as Kolmogorov-Arnold theorem ensures that ANNs can be used as an unbiased estimator of data. They performed the analysis of the form factor shapes using a system of artificial neural networks trained from experimental pseudodata and averaged together to predict form factor shapes with a prescribed uncertainty. This technique will be used for computing nuclear matrix elements in Sect. IV H.

Theoretical predictions for conversion rates in such processes contain uncertainties due to non-perturbative QCD effects in single and multi-nucleon sector. The operators of the Standard Model Effective Field Theory, written in terms of quark fields, need to be matched to the effective operators written in terms of the nucleon degrees of freedom. M.R. Schindler has extensive experience applying EFT methods to compute single- and multi-nucleon processes.

Theoretical predictions for conversion rates in muon conversion processes contain uncertainties due to nuclear binding effects. Accurate theoretical predictions of the nuclear are essential to realize the full potential of these experiments. Currently, such effects are estimated with the help of phenomenological nuclear models. The USC particle and nuclear theory group members (V. Gudkov) have extensive experience computing hadronic and nuclear effects and will provide required computations and numerical simulations.

D.J. Tedeschi is currently a member of the computing infrastructure group for the Mu2e collaboration. His responsibilities include developing the data quality monitoring components for Mu2e data production. Tedeschi's recent work with the Majorana Demonstrator Project (MJD) investigated neutrinoless double beta decay [20] as well as axion and exotic dark matter searches.[21, 22] As a member of the Majorana Collaboration, D. Tedeschi took a lead role in data production, managing data flow from the experimental site at Sanford Underground Research Facility (SURF) to the analysis center at the National Energy Research Scientific Computing Center (NERSC); as well as the transformation of that data (production) into files compatible with collaboration analysis tools. D. Tedeschi was responsible for creating the blind analysis software framework for the neutrinoless double beta decay measurement as well as for analysis of the scintillator veto system. Prior to working on MJD, D. Tedeschi's efforts at Thomas Jefferson National Accelerator Facility were focused on photonuclear physics using the CLAS spectrometer. As a member of the CLAS collaboration, Tedeschi played a lead role in the construction of CLAS drift chambers[23, 24], development of a polarized target magnet[25], and photon beam-line devices[26, 27]. He made significant contributions on physics publications involving the search for pentaquark states[28], tests of the vector dominance model (VDM) in ϕ -meson production on light nuclei[29–31], and the role of nuclear medium modifications.[32]

The discussion above shows that the PIs have the relevant experience needed for the successful execution of the proposed activities described below. The projects will also involve several graduate students and postdoctoral scholars working closely with the Mu2e Collaboration. Currently D.J Tedeschi (senior member), graduate student A. Boldy (regular member), A. Petrov (theorist member), and Girish Kumar (theorist postdoc) are official members of the Mu2e Collaboration. (See letter of support from R. Bernstein and S. Miscetti.)

B. Textbooks, scientific service and outreach

Petrov published a textbook “Indirect Searches for New Physics” (CRC Press) in 2021 [5]. This book discusses searches for NP with charged leptons, neutrinos, and quarks, emphasizing a distinction between model-dependent and model-independent methods employed to make predictions. This book follows a textbook “Effective Field Theories” (World Scientific) [2] by Petrov and Blechman published in 2015. Petrov was involved in the writing and editing of several important review papers. Among those is the “Report of the Frontier For Rare Processes and Precision Measurements” [17], which Petrov wrote with his co-conveners, M. Artuso and R. Bernstein, as well as a number of Contributed Papers submitted to the Snowmass21 [16].

Petrov has maintained a high visibility in the physics community. He was a co-convenor of the Frontier for Rare Processes and Precision Measurements (RPF) of the Snowmass 2021 [33] (with M. Artuso and R. Bernstein). Petrov serves as a member of the Belle Program Advisory

Committee (BPAC), Japan’s High Energy Accelerator Research Organization (KEK) (2020-23), providing recommendations on scientific issues related to the KEK’s B-factory project. Petrov served on the IFF Award Committee at Fermilab for 2016-20. Petrov serves as a convener of “Beyond the SM” topical group of the International Quarkonium Working Group (QWG) [34], an organization of physicists working on problems associated with the physics of quarkonia. In 2024, Petrov joined the Mu2e collaboration as a theory member.

In the last four years, Petrov has helped organize several conferences. As part of his duty as the Snowmass 2021 Frontier convener, he co-organized several workshops on topics related to the Frontier for Rare Processes and Precision Measurements. Petrov was a member of the International Advisory Committee for CHARM workshops in Mexico City in 2020 and Siegen in 2023, as well as the 20th Conference on Flavor Physics and CP Violation (FPCP 2022), and International Workshop on the CKM Unitarity Triangle (CKM 2021). Petrov helped to organize QWG workshops at UC Davis (2021) and GSI (2022), as well as previous workshops in this series. He recently chaired the local organizing committee for the International Conference on Beauty, Charm, and Hyperons (BEACH 2024), which was held in Charleston, SC.

IV. PROPOSED PROJECTS AND RELEVANT ANALYSES

One of the most important classes of uncertainties in muon conversion experiments includes physical backgrounds. Such backgrounds can be reduced by computing the relevant physical processes. We propose to work closely with the Mu2e collaboration to provide theoretical computations needed to reduce such backgrounds. The setup of the Mu2e and follow-up Fermilab experiments (Mu2e-II/AMF) is “charge symmetric,” so studies of both $\mu^- + (A, Z) \rightarrow e^- + (A, Z)$ and $\mu^- + (A, Z) \rightarrow e^+ + (A, Z - 2)$ processes are possible with comparable detection efficiencies for electrons and positrons. Compared to the electron energy in $\mu^- \rightarrow e^-$ conversion, the latter process results in much lower positron energy, which populates the energy regions populated by poorly understood Standard Model backgrounds. Proper evaluation of such backgrounds is of the utmost importance for the Mu2e experiment. We propose theoretical computation of the effects listed below, which would then be adapted to the Mu2e simulation framework to account for the novel experimental acceptance of the Mu2e detector system. Such close coordination of theory and experiment is crucial to understanding the results of NP searches.

A. TIME collaboration and collaboration meetings

One of the centerpieces of this proposal is the formation of the collaboration of theorists working closely with the Mu2e collaboration on physics and background computations and implementations of those in the Monte Carlo generators deployed by Mu2e. We envision the formation of the “Mu2e theory group” based at the University of South Carolina (USC) but including physicists from other US Universities. The new theory-experiment collaboration “Theory for Intensity-frontier Muon Experiments (TIME)” would include PI/co-PIs for this proposal, Mu2e Collaboration members, and external theorists. Some of those theorists already expressed interest in the project: see enclosed letters of collaboration from B. Bernstein, V. Cirigliano, W. Haxton, R. Plestid, E. Rule, and J. Zupan, who, along with the PIs will form the core of TIME collaboration. These physicists will bring their unique knowledge to USC, building intellectual infrastructure lasting long past the successful completion of activities described in this proposal, making USC the focal point of theoretical activities related to future muon experiments at FNAL

(Mu2e-II, Advanced Muon Facility (AMF), and potentially, muon collider) and other national and international labs.

The theorists forming TIME collaboration have complementary research interests relevant to muon conversion physics, which means that they will take leadership in various aspects of the proposed activities, such as QCD effective field theories for muon conversion (W. Haxton, E. Rule, J. Zupan), EFTs and new physics analyses (J. Zupan), radiative muon conversion backgrounds (R. Plestid), EFTs and nuclear analyses (V. Cirigliano). They complement PIs' research interests in nuclear transition calculations (V. Gudkov), QCD EFTs and flavor physics (A. Petrov), nucleon EFTs and large- N_c (M. Schindler). Their expertise will be invaluable for the Monte Carlo simulation codes created by the experimental members of the TIME collaboration, both at USC (D. Tedeschi) and elsewhere (R. Bernstein).

We are planning focused two yearly progress meetings at USC to discuss the progress with the computations, new problems and opportunities, and implementation. In addition to the scientific interchange, we will use the meetings as training opportunities in scientific presentations for our postdocs and graduate students. We request \$30K/year in conference support and \$40K/year in travel funds to support such meetings and other related travel expenses.

B. Theoretical studies of experimental backgrounds: RMC

Radiative muon capture (RMC) is an intrinsic background at Mu2e and Mu2e-II $\mu^- + (A, Z) \rightarrow \nu + (A, Z - 1) + \gamma$, where γ converts into the e^+e^- pair. Such conversion can happen inside the target nucleus (and with virtual photons) and can mimic the signal. The maximum photon energy kinematically allowed is dependent on the nuclear target [35]. While the nucleon-level process $p + \mu^- \rightarrow n + \nu + \gamma$ has been computed and measured (at TRIUMF), the theoretical prediction of the rate of this process for muon conversion experiments on nuclei requires much more sophisticated theoretical techniques [36]. Currently, the closure approximation (CA) is employed, in which one assigns a single average excitation energy to the final nucleus, implying a single maximum photon energy. This simple model is used to predict the total rates and the bulk of the spectrum shape, not to model the high-momentum region accurately. The RMC background is highly sensitive to the choice of the nuclear target, as shown by TIME collaboration member R. Plestid in [37]. We propose to go beyond the CA approximation and predict RMC spectra for both e^+ and e^- for several different target nuclei (including Al , Tl , and Au).

It is interesting to point out that the most recent available data from the SINDRUM II experiment, which used Au as a target, indicated a possible excess in the tail of the positron spectrum. This excess is not explained by the calculations made with the closure approximation. The published SINDRUM II data are in significant tension with the current computations of the RMC effect, making it impossible to distinguish between SM and possible NP contributions. Notably, the RMC process can also be generated by the NP operators that differ from those in Eq. (1). The USC theory group (V. Gudkov, A. Petrov, a postdoc and graduate students) proposes to evaluate those contributions, computing the RMC spectrum for electrons and positrons for several different nuclei in the Mu2e/Mu2e-II/AMF setup. The TIME collaboration members W. Haxton, R. Plestid, and E. Rule will be involved in this computation. The USC experimental effort (D. Tedeschi, post doc, graduate student) will process the new calculations using the Mu2e simulation framework to see the effects on the measured e^- and e^+ momentum spectra.

C. Studies of theoretical uncertainties related to nuclear effects

Charged lepton flavor-violating processes offer interesting possibilities for probing new physics at multi-TeV scales. Any NP scenario that involves LFV interactions can be matched to an effective Lagrangian, $\mathcal{L}_{\text{eff}}$, whose Wilson coefficients would be determined by the ultraviolet (UV) physics that becomes active at some scale Λ . The Lagrangian that is relevant for muon conversion was introduced in Sect. II. That effective Lagrangian, however, is written in terms of quark-level operators, whereas the relevant degrees of freedom for muon conversion are nucleons and nuclei [38–40]. One must rewrite $\mathcal{L}_{\text{eff}}$ in terms of the relevant degrees of freedom. This is usually achieved by matching the Lagrangian first to the one that involves protons and neutrons, as was shown by TIME collaboration member V. Cirigliano in [41],

$$\begin{aligned} m_q \bar{q}q &\rightarrow f_{SN}^{(q)} m_N \bar{\psi}_N \psi_N \\ \bar{q}\gamma_\rho q &\rightarrow f_{VN}^{(q)} \bar{\psi}_N \gamma_\rho \psi_N \\ \frac{\beta_L}{2g_s^3} GG &\rightarrow f_{GN} m_N \bar{\psi}_N \psi_N . \end{aligned} \quad (4)$$

where the index $N = p, n$ represents the proton or a neutron, and ψ_N represent the nucleon interpolating fields. The f 's are the nucleon form factors, which, in principle, depend on the momentum transfer.

The well-known relation involving the trace of the energy-momentum tensor of QCD, $\Theta_{\mu\nu}$, which states $\langle N | \Theta_\mu^\mu | N \rangle = m_N \langle N | \bar{\psi}_N \psi_N | N \rangle$, can be used to eliminate the gluon form-factor f_{GN} in terms of the scalar nucleon form factors $f_{SN}^{(q)}$,

$$f_{GN} + \sum_{q=u,d,s} f_{SN}^{(q)} = 1 \quad (5)$$

The scalar nucleon form factors, however, cannot be obtained from symmetry arguments and must be computed, with their computation being non-trivial. Alternatively, one can keep the gluon form factor and eliminate one of the $f_{SN}^{(q)}$. We'll do so in what follows.

Computing the matching coefficients of the nucleon-level Lagrangian then makes it possible to take the relevant nuclear-level matrix elements,

$$\begin{aligned} \langle A, Z | \bar{\psi}_p \psi_p | A, Z \rangle &= Z \rho^{(p)} \\ \langle A, Z | \bar{\psi}_n \psi_n | A, Z \rangle &= (A - Z) \rho^{(n)} \\ \langle A, Z | \bar{\psi}_p \gamma^0 \psi_p | A, Z \rangle &= Z \rho^{(p)} \\ \langle A, Z | \bar{\psi}_n \gamma^0 \psi_n | A, Z \rangle &= (A - Z) \rho^{(n)} , \end{aligned} \quad (6)$$

where $\langle A, Z | \dots | A, Z \rangle$ represents the matrix element between the ground state of the target nuclei with $\rho^{(p)}$ and $\rho^{(n)}$ parameterizing the proton and neutron densities. Note that because of the small recoil, the space-like matrix elements are assumed to be zero, $\langle A, Z | \bar{\psi}_N \gamma^i \psi_N | A, Z \rangle = 0$. The final result is rather cumbersome [8] and depends on the so-called overlap integrals $D, V^{(N)}, S^{(N)}$ which are obtained from the overlap of the muon and electron wave functions weighted by the nucleon densities. Such theoretical predictions for conversion rates in such processes contain uncertainties due to *nuclear* binding effects. Currently, such effects are estimated with the help of phenomenological nuclear models [8]. It appears from the Fig. 1 that the effects of some quark operators can be separated by following their target (Z) dependence. The USC theorists (V. Gudkov, A. Petrov, a postdoc and graduate students) proposes to test this by

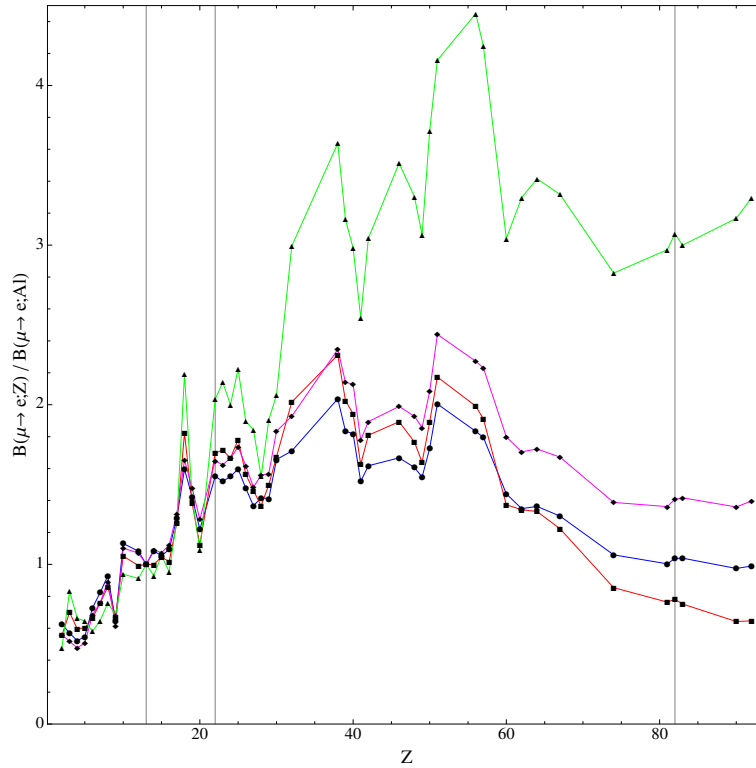


FIG. 1. Target Z dependence of the conversion rate (from [41]) in the single-operator dominance hypothesis.

computing the form factors and overlap integrals above with several nuclear models (such as the Correlated Fermi Gas Model [42]) and use the spread of the results for each Z as an estimate of the nuclear uncertainty.

The computations of the positron spectrum in $\mu^- + (A, Z) \rightarrow e^+ + (A, Z - 2)$ transitions are usually done, assuming that the transitions happen between the ground states of the target and the final state daughter nuclei. For example, in the case of the Mu2e experiment, the target nucleus is ^{27}Al , while the final state nucleus is ^{27}Mg . This, however, does not need to be the case. In this case, the transitions to the non-ground state configurations would affect the resulting spectrum of the positrons. V. Gudkov, A. Petrov, a postdoc and a graduate student propose to evaluate the effect of those transitions on the positron spectra for different target nuclei, collaborating with TIME collaboration members V. Cirigliano, E. Rule, and W. Haxton.

D. Gluonic Rayleigh operators and Mu2e

Instead of removing the gluon form factor, it can be calculated [11]. The initial state in the μ - e conversion process

$$\mu^- + (A, Z) \rightarrow e^- + (A, Z) \quad (7)$$

is the $1s$ state of the muonic atom with the binding energy E_b , and the final electron state is the eigenstate with the energy $m_\mu - E_b$. Following Ref. [8], the μ - e conversion amplitude can

be written as

$$M_{\ell gg}^{\mu e} = \frac{1}{\Lambda^2} \int d^3x \left(c_1 \bar{\psi}_{\kappa, W}^{\mu(e)} P_L \psi_{1s}^{(\mu)} + c_3 \bar{\psi}_{\kappa, W}^{\mu(e)} P_R \psi_{1s}^{(\mu)} \right) \langle A, Z | \frac{\beta_L}{4\alpha_s} G_{\mu\nu}^a G^{a\mu\nu} | A, Z \rangle \quad (8)$$

where $\langle A, Z |$ and $| A, Z \rangle$ are the final and initial states of the nucleus, respectively. Here c_i are the combinations of the Wilson coefficients in Eq. (3), and the $1s$ initial muon wave function $\psi_{1s}^{(\mu)}$ is normalized to one [11].

The matrix element in Eq. (8) relevant to the coherent conversion process $\mu^- + (A, Z) \rightarrow e^- + (A, Z)$ can be expressed by the proton $\rho^{(p)}$ and neutron $\rho^{(n)}$ densities in a nucleus as

$$\langle A, Z | \frac{\beta_L}{4\alpha_s} G_{\mu\nu}^a G^{a\mu\nu} | A, Z \rangle = -\frac{9}{2} \left[Z G^{(g,p)} \rho^{(p)} + (A - Z) G^{(g,n)} \rho^{(n)} \right], \quad (9)$$

where A and Z again represent the mass and atomic number of the nucleus and the matrix element of the gluon operator between the nucleon states is defined as

$$G^{(g, \mathcal{N})} = \langle \mathcal{N} | \frac{\alpha_s}{4\pi} G_{\mu\nu}^a G^{a\mu\nu} | \mathcal{N} \rangle \quad (10)$$

with $\mathcal{N} = n, p$. This scalar matrix element can be calculated by using the trace of the energy-momentum tensor Θ_μ^μ and applying the flavor $SU(3)$ symmetry [11]. Since Θ_μ^μ is flavor symmetric, the proton and neutron scalar matrix elements are equal. For the strange-quark sigma term $\sigma_s \equiv m_s \langle p | \bar{s}s | p \rangle = 50$ MeV the numerical result is $G^{(g, \mathcal{N})} = -189$ MeV [11].

The nucleon densities are assumed spherically symmetric and normalized as

$$\int_0^\infty dr 4\pi r^2 \rho^{(\mathcal{N})}(r) = 1, \quad \text{with} \quad \rho(r) = \frac{\rho_0}{1 + \exp[(r - c)/z]}, \quad (11)$$

where, as usual, for the spherical nuclei, the two-parameter Fermi (2pF) charge distribution is used with c as the half-density radius. We can use different parametrizations to see the model dependence of the results. The formula for the coherent conversion rate, assuming the dominance of the gluon Rayleigh operators can be written as

$$\Gamma_{\text{conv}}(\mu N \rightarrow e N) = \frac{4}{\Lambda^4} (|c_1|^2 + |c_3|^2) a_N^2, \quad (12)$$

where c_i are proportional to C_{GR} and C_{GL} , and

$$a_N = G^{(g,p)} S^{(p)} + G^{(g,n)} S^{(n)}. \quad (13)$$

The overlap integrals are defined as [8]

$$\begin{aligned} S^{(p)} &= \frac{1}{2\sqrt{2}} \int_0^\infty dr r^2 Z \rho^{(p)} (g_e^- g_\mu^- - f_e^- f_\mu^-), \\ S^{(n)} &= \frac{1}{2\sqrt{2}} \int_0^\infty dr r^2 (A - Z) \rho^{(n)} (g_e^- g_\mu^- - f_e^- f_\mu^-). \end{aligned} \quad (14)$$

These overlap integrals can be evaluated with the help of nuclear models [11] or using ab initio techniques [43].

Here, Petrov proposes yet another alternative way to estimate the matrix elements of the gluonic operators, which relies more on other experimental measurements. The idea is as follows.

The mass-radius of the proton can be obtained through the measurement of its gravitational scalar form factor [44]. This form factor can be obtained by studying the threshold production of a J/ψ meson on the proton using the data obtained at the Jefferson Lab experiments, for example, GlueX. The matrix element of the process $\gamma + p \rightarrow J/\psi + p$ can be written in terms of the matrix element in Eq. (10) [44],

$$\mathcal{M}_{\gamma p \rightarrow J/\psi p} = -Qec_2 \frac{16\pi^2 m_p}{b} \langle p | G_{\mu\nu}^a G^{a\mu\nu} | p \rangle, \quad (15)$$

where the definitions of various parameters in Eq. (15) can be found in [44]. Integrating over t , the momentum transfer yields the cross-section for the threshold J/ψ production on a proton $\sigma_{\gamma p \rightarrow J/\psi p}(s)$. It is interesting to note that the gluonic matrix element in Eq. (15) is exactly what is needed for evaluating Eq. (10), so it can be extracted from the measured value of the J/ψ threshold production cross-section.

One can extend this idea if the *nuclear* threshold production cross sections are available for the nuclei that are used as targets for the Mu2e experiment, i.e., if the nuclear matrix elements $\langle A, Z | \frac{\beta_L}{4\alpha_s} G_{\mu\nu}^a G^{a\mu\nu} | A, Z \rangle$ for aluminum or gold can be extracted directly. Preliminary discussions with the USC nuclear experimentalists (Prof. Pawel Nadel-Turonski) indicate that such measurements could be done at the Electron-Ion Collider (EIC). Alternatively, relevant data can be obtained from ultra-peripheral collisions of nuclei at ALICE or even JLab experiments. As part of this proposal, Petrov will provide a framework for the extraction of nuclear gluonic matrix elements.

The computations of the positron spectrum in $\mu^- + (A, Z) \rightarrow e^+ + (A, Z - 2)$ transitions are usually done, assuming that the transitions happen between the ground states of the target and the final state daughter nuclei. For example, in the case of the Mu2e experiment, the target nucleus is ^{27}Al , while the final state nucleus is ^{27}Mg . This, however, does not need to be the case. In this case, the transitions to the non-ground state configurations would affect the resulting spectrum of the positrons. We (V. Gudkov, postdoc, and graduate student) propose to evaluate the effect of those transitions on the positron spectra for different target nuclei. This project will be assisted by TIME collaboration members W. Haxton and E. Rule.

E. Photonic Rayleigh operators and Mu2e

Besides contributions from dimension-six operators of Eq. (1) described in [8], other, formally higher-dimensional operators of the Rayleigh type, $\mathcal{L}_R$, driving $\mu e \gamma \gamma$ interactions, contribute to muon conversion.

$$\mathcal{L}_{\ell\gamma\gamma} = \frac{\alpha}{12\pi} \frac{m_\mu}{\Lambda^2 v} (C_{FF}^L \bar{\mu} P_L e + C_{FF}^R \bar{\mu} P_R e) F_{\mu\nu} F^{\mu\nu} + h.c. \quad (16)$$

where v is the Higgs vacuum expectation value, $C_{FF}^{L/R}$ are the Wilson coefficients, and $F_{\mu\nu}$ is electromagnetic strength tensor. It has been shown that such operators can be suppressed by only two powers of the NP scale Λ [11, 45], so their effects are not necessarily small compared to dim-6 operators considered in [8]. Analyses of their contributions to μ conversion involve the amplitude

$$M_{\mu \rightarrow e} = \frac{1}{\Lambda^2} \int d^3 r \bar{\psi}_e(\vec{r}) (C_{FF}^L P_L + C_{FF}^R P_R) \psi_\mu^{1s}(\vec{r}) \langle A, Z | F_{\mu\nu} F^{\mu\nu} | A, Z \rangle, \quad (17)$$

where $\psi_\mu^{1s}(\vec{r})$ and $\psi_e(\vec{r})$ are the wave functions of a $1s$ bound muon and a produced electron. Petrov, a postdoc and a graduate student will revisit the contribution of $\mu \rightarrow e \gamma \gamma$ to muon conversion. The matrix elements in Eq. (17), $\langle A | F_{\mu\nu} F^{\mu\nu} | A \rangle$, although for the different nuclei,

have been studied for Dark Matter experiments looking for Magnetic Inelastic Dark Matter [46, 47]. They also define the cross sections of non-relativistic *scalar* DM ϕ , which interacts with visible matter through Rayleigh operators $\sim \phi^2 F^2$ [48], which could explain the excess of “electron recoil” events recently observed by the Xenon1T collaboration (see sect. II). Petrov (together with J. Zupan and other members of TIME collaboration) proposes to compute the matrix elements $\langle A, Z | F_{\mu\nu} F^{\mu\nu} | A, Z \rangle$, including the contributions of two-nucleon correlations [49] to determine the contribution of $\mu \rightarrow e\gamma\gamma$ process to muon conversion.

F. Studies of theoretical uncertainties related to nucleons

It was shown that various NP models can be discriminated in $\mu^- + (A, Z) \rightarrow e^- + (A, Z)$ transitions by comparing the conversion rates on different nuclei [38, 50]. By adapting EFT methods developed by Rule, Haxton, and collaborators, we (A. Petrov and M. Schindler together with collaborators from the TIME collaboration) propose to study the sensitivity of $\mu^- + (A, Z) \rightarrow e^+ + (A, Z - 2)$ processes on various high-Z targets to various NP models. For Majorana neutrino exchange, the hadronic structure of the operators contributing to $\mu^- + (A, Z) \rightarrow e^+ + (A, Z - 2)$ is analogous to that of operators contributing to neutrinoless double beta decay [36]. These operators have been studied in detailed, see e.g., Ref. [51]. However, the nuclear targets considered for $\mu^- \rightarrow e^+$ are different from those in neutrinoless double beta decay. We also plan to study possible NP mechanisms beyond Majorana neutrino exchange, their manifestation at the hadronic level, and their impact on $\mu^- \rightarrow e^+$ in target nuclei.

G. Probing unconventional NP models: baryon number violation (BNV)

One of Sakharov’s conditions for matter-antimatter asymmetry of the Universe requires baryon or lepton number violation. Classically, baryon (B) and lepton (L) numbers are conserved in the Standard Model, requiring a complicated symmetry-breaking mechanism to be realized in the early Universe. This is not necessarily true in theories beyond the Standard Model. There are explicit models where proton decay is forbidden or suppressed, but processes like $\Delta B = \Delta L = 2$ are allowed to proceed [52, 53] even though they are mediated by higher-dimensional operators. Examples of such processes are neutron-antineutron $n - \bar{n}$, or hydrogen-antihydrogen $H - \bar{H}$ oscillations. These processes, e.g., $H - \bar{H}$ oscillations, have been studied theoretically and experimentally, with experimental constraints obtained from astrophysical observations, employing stability of matter arguments or data from neutrino experiments, where large fiducial volumes of the detectors allow to search for very rare processes even with finite exposure time.

A useful observation is that the $H - \bar{H}$ mixing is driven by the underlying $e^- p \rightarrow e^+ \bar{p}$ process, which can also be probed in multi-electron atoms. The resulting predictions will depend on the sizes of relevant nuclear matrix elements, which usually constitute theoretical uncertainty. These matrix elements will be evaluated using nuclear models. Interestingly, these uncertainties are similar to those affecting $\mu^- \rightarrow e^+$ conversion rates to be studied at Fermilab’s Mu2e experiment along with the $\mu^- \rightarrow e^-$ conversion process. This process, $\mu^- p \rightarrow e^+ \bar{p}$, initiated by a muon, contributes to the second most important LFV signature to be studied by the Mu2e collaboration, $\mu^- + (A, Z) \rightarrow e^+ + (A, Z - 2)$! Using the computational techniques developed above, we (Petrov, Tedeschi) will compute the BNV contribution to this experimental signature and constrain the Wilson coefficients of the corresponding operator from the Mu2e/COMET experimental data.

H. Nuclear effects and machine learning (ML)

As was pointed out above, the results of the analysis of muon conversion $\mu^- + (A, Z) \rightarrow e^- + (A, Z)$ depend on nucleon densities inside the nuclei [8]. It is a challenging task to understand the universal behavior of nucleons inside a diverse set of nuclei employed in those experiments. Ultimately, such matrix elements will be studied in ab-initio computations [43].

Petrov and a USC graduate student (S. Mohammad) will take a different approach. Instead of finding a universal (distribution) function with variable parameters that describes the motion of protons and neutrons inside all nuclei, they will look at each (A, Z) nuclei individually. They will approximate the relevant nuclear densities by employing methods of machine learning. In particular, they will use artificial neural networks (ANNs), exploring their ability to provide unbiased universal approximants to incomplete data, building a statistical interpolating model of nucleon momentum distribution functions based on ANNs that contains information on experimental uncertainties and correlations but does not introduce theoretical bias. These ANNs will be trained on experimental data from electron or neutrino flavor-conserving scattering experiments on those nuclei. While novel, this technique has already been tested: Petrov and students used this method to study semileptonic form factors in D -decays [19]. It was also used to study parton distribution functions [54]. Such analyses could also be beneficial for the understanding of SM neutrino-nucleus interactions, which is important for current and upcoming neutrino experiments, including DUNE. This study would enable to start of the *long-term program* of investigation into applications of ML to studies of nuclear uncertainties in low-energy experiments extracting the structure of the fundamental Lagrangian from experimental data.

I. Mu2e Experiment Simulation

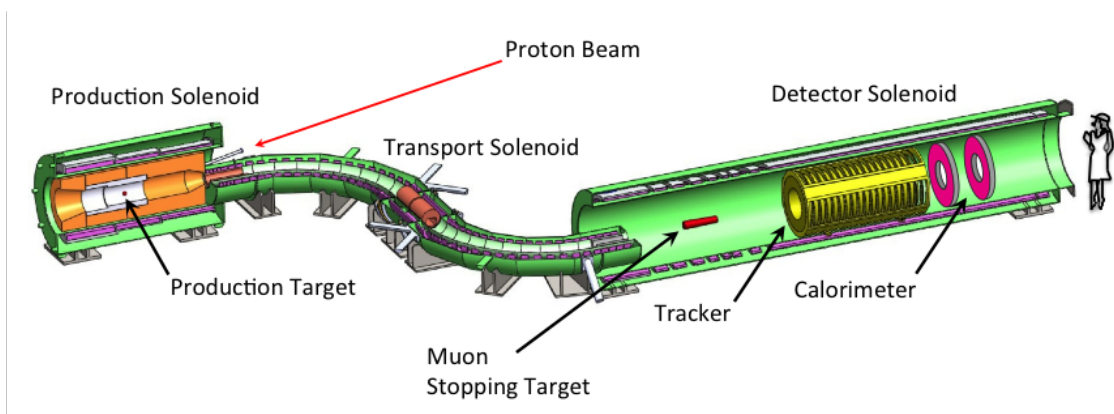


FIG. 2. Representation of the Mu2e detector system.

The Mu2e project aims to identify electrons and positrons from muons stopped in the aluminum target by detecting their momentum and energy using a straw-tube tracker in a solenoidal magnetic field and electromagnetic calorimeter as shown in the figure.[55] The muons are created by an intense, pulsed proton beam incident on a high-Z production target. Backward going electrons, muons, and pions are transported through the chicane via the transport solenoid where they then interact with the stopping target. The annular design of the detector system is designed to reduce the exposure to beam-related backgrounds. The use of a pulsed-beam structure allows timing cuts to remove the background and isolate muons captured by the Aluminum nuclei in the stopping target. This unique shape, however, creates a non-uniform acceptance that must

accounted for by the Mu2e simulation programs. Theoretical predictions must be put through this simulation system to produce realist comparisons to the measured data. This argument is made clear by examining the acceptance (a) function for a detector to a given theoretical amplitude $D(x)$ for all detected events (Q),

$$a = \int_Q E(x)D(x) d^m x. \quad (18)$$

The efficiency of the Mu2e detector, i.e. the probability of detecting an event of parameters x , is represented by $E(x)$. This quantity is determined by geometric models of the detector and the solution of the integral is achieved via Monte Carlo methods over the entire phase space (Q).

Individual physics process contributions such as cosmic rays (CORSIKA), continuous electrons spectra, radiative muon decay, muon decay in orbit (DIO), etc... as well as beam backgrounds from the beam flash are produced and mixed together as input to the GEANT simulation code. Detection efficiencies are accounted for as detector hits are recorded and digitized to create an output data stream for analysis. The USC experimental group (Tedeschi, postdoc, and students) will perform simulations with both the collaboration MC input as well as the MC inputs developed in this proposal for comparison.

V. PERSONNEL AND PROPOSED TIMELINE OF ACTIVITIES

We request funding of salaries for the personnel involved in the project: three graduate students and postdoctoral associates. We also request summer salary support for the senior personnel involved in the project: Prof. A. Petrov (1 month), Prof. V. Gudkov (2 months), Prof. M. Schindler (1 month), and Prof. D.J. Tedeschi (2 months). We will be working with other theorists at USC with similar research interests, as well as with current theory graduate students (S. Muhammad) and postdocs (Dr. G. Kumar) in the Department. Support for an additional postdoc and graduate student in experimental physics will be used for the translation of the theoretical work into the Mu2e analysis framework. Recognizing the importance of the distribution of the researchers at various career stages for this project, we will involve senior undergraduate students participating in the newly approved USC Research Experience for Undergraduates (REU) program in the projects described in this proposal. Below is the timeline of the proposed activities and the responsible personnel.

Year 1

If the proposal is selected for funding, we will immediately start the formation of the "Theory for Intensity-frontier Muon Experiments" (TIME) collaboration and hold a collaboration meeting (responsible: A. Petrov), which would include PI/co-PIs for this proposal, other USC faculty, Mu2e Collaborators, and core external theorists (see enclosed letters of collaboration from B. Bernstein (FNAL), V. Cirigliano (INT Seattle), W. Haxton (UC Berkeley), R. Plestid (Caltech), E. Rule (UC Berkeley), and J. Zupan (Cincinnati)). We emphasize that no direct support is requested for external collaborators, but they will be participating in research and collaboration meetings hosted by USC. All activities described below will be done together with researchers constituting TIME collaboration.

We will immediately start searching for postdoctoral research associates with experience relevant for the computations of the RMC effect in nucleons and nuclei (responsible: A. Petrov). We will also start recruiting graduate students.

The collaboration will start computations of the RMC effect (responsible: Petrov, Gudkov, postdoc, students, TIME Collaboration) in $\mu^- + (A, Z) \rightarrow e^+ + (A, Z - 2)$. We will compute nuclear effects related to the possible $\mu \rightarrow e\gamma\gamma$ contribution to $\mu^- + (A, Z) \rightarrow e^- + (A, Z)$ (responsible: Petrov and students).

We will evaluate the non-ground state contribution to the transition between the target and the final state daughter nuclei in $\mu^- + (A, Z) \rightarrow e^- + (A, Z)$ (responsible: V. Gudkov, postdoc, student).

We will start the computation of the contribution of baryon-number violating operators to $\mu^- + (A, Z) \rightarrow e^+ + (A, Z - 2)$ (responsible: Petrov, postdoc, students).

We will start the computation of $\mu^- + (A, Z) \rightarrow e^+ + (A, Z - 2)$ conversion rates for different nuclei (responsible: Petrov and Schindler).

In order to process the new calculations through the Mu2e detector system, we will produce working Monte Carlo analysis codes that are compatible with the Mu2e simulation and analysis software (responsible: Tedeschi, experimental postdoc, graduate student A. Boldy).

Year 2

Finish computation of the RMC effect in $\mu^- + (A, Z) \rightarrow e^+(A, Z - 2)$ (responsible: Petrov, Gudkov, postdoc, students, TIME Collaboration), and start embedding the results into the Mu2e Monte Carlo generators (responsible: Petrov, Tedeschi, postdocs, graduate students, TIME Collaboration).

We will continue computing of the contribution of baryon-number violating operators to $\mu^- + (A, Z) \rightarrow e^+ + (A, Z - 2)$ (responsible: Petrov, Tedeschi, postdocs, graduate students).

Determine matrix elements contributing to $\mu^- + (A, Z) \rightarrow e^+(A, Z - 2)$ for operators originating from Majorana neutrino exchange (responsible: Petrov and Schindler with TIME collaborators).

Produce trained artificial neural net parameterizing nuclear effects in $\mu^- + (A, Z) \rightarrow e^- + (A, Z)$ and $\mu^- + (A, Z) \rightarrow e^+ + (A, Z - 2)$ (responsible: Petrov, postdoc, TIME Collaboration).

We envision for the TIME collaboration to play a leading role in muon-physics related theoretical computations after the completion of this project. Physics background computations and their simulations for Mu2e-II/AMF and muon collider will be needed long past 2027.

VI. SUMMARY

The proposed research program seeks to establish close collaboration between Mu2e/Advanced Muon Facility experimentalists and USC particle theory to compute the needed SM background contributions affecting signal interpretation in the Mu2e/Mu2e-II/AMF experiments. It is strongly aligned with the 2024 recommendations of the Particle Physics Project Prioritization Panel and the 2021-22 Snowmass study on the future directions for US high energy physics. These efforts will be instrumental in elevating the level of scientific discourse at USC by attracting leading experts in the field to the TIME collaboration. The exposure to world class scientists and programs this grant supports will provide USC postdoctoral associates and graduate students with the highest level of scientific training.

APPENDIX 1: BIBLIOGRAPHY & REFERENCES CITED

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- [1] A. Czarnecki, W. J. Marciano, and K. Melnikov, “Coherent muon electron conversion in muonic atoms,” *AIP Conf. Proc.* **435** no. 1, (1998) 409–418, [arXiv:hep-ph/9801218](#).
 - [2] A. A. Petrov and A. E. Blechman, *Effective Field Theories*. World Scientific, 2016. <http://www.worldscientific.com/worldscibooks/10.1142/8619>.
 - [3] D. E. Hazard and A. A. Petrov, “Lepton flavor violating quarkonium decays,” *Phys. Rev.* **D94** no. 7, (2016) 074023, [arXiv:1607.00815 \[hep-ph\]](#).
 - [4] R. H. Bernstein and P. S. Cooper, “Charged Lepton Flavor Violation: An Experimenter’s Guide,” *Phys. Rept.* **532** (2013) 27–64, [arXiv:1307.5787 \[hep-ex\]](#).
 - [5] A. A. Petrov, *Indirect Searches for New Physics*. CRC Press, Boca Raton, 5, 2021. <https://www.routledge.com/Indirect-Searches-for-New-Physics/Petrov/p/book/9780815386049>.
 - [6] **Super-Kamiokande** Collaboration, S. Sussman *et al.*, “Dinucleon and Nucleon Decay to Two-Body Final States with no Hadrons in Super-Kamiokande,” [arXiv:1811.12430 \[hep-ex\]](#).
 - [7] R. J. Hill, R. Plestid, and J. Zupan, “Searching for new physics at $\mu \rightarrow e$ facilities with μ^+ and π^+ decays at rest,” [arXiv:2310.00043 \[hep-ph\]](#).
 - [8] R. Kitano, M. Koike, and Y. Okada, “Detailed calculation of lepton flavor violating muon electron conversion rate for various nuclei,” *Phys. Rev. D* **66** (2002) 096002, [arXiv:hep-ph/0203110](#). [Erratum: *Phys.Rev.D* 76, 059902 (2007)].
 - [9] S. L. Glashow, D. Guadagnoli, and K. Lane, “Lepton Flavor Violation in B Decays?,” *Phys. Rev. Lett.* **114** (2015) 091801, [arXiv:1411.0565 \[hep-ph\]](#).
 - [10] D. E. Hazard and A. A. Petrov, “Radiative lepton flavor violating B , D , and K decays,” *Phys. Rev. D* **98** no. 1, (2018) 015027, [arXiv:1711.05314 \[hep-ph\]](#).
 - [11] A. A. Petrov and D. V. Zhuridov, “Lepton flavor-violating transitions in effective field theory and gluonic operators,” *Phys. Rev. D* **89** no. 3, (2014) 033005, [arXiv:1308.6561 \[hep-ph\]](#).
 - [12] K. J. Healey, A. A. Petrov, and D. Zhuridov, “Nonstandard neutrino interactions and transition magnetic moments,” *Phys. Rev. D* **87** no. 11, (2013) 117301, [arXiv:1305.0584 \[hep-ph\]](#). [Erratum: *Phys.Rev.D* 89, 059904 (2014)].
 - [13] K. P. Jungmann, “Precision Muonium Spectroscopy,” *J. Phys. Soc. Jap.* **85** no. 9, (2016) 091004, [arXiv:1603.01195 \[hep-ex\]](#).
 - [14] R. Conlin and A. A. Petrov, “Muonium-antimuonium oscillations in effective field theory,” *Phys. Rev. D* **102** no. 9, (2020) 095001, [arXiv:2005.10276 \[hep-ph\]](#).
 - [15] A. A. Petrov, R. Conlin, and C. Grant, “Studying $\Delta L = 2$ Lepton Flavor Violation with Muons,” *Universe* **8** no. 3, (2022) 169, [arXiv:2203.04161 \[hep-ph\]](#).
 - [16] A.-Y. Bai *et al.*, “Snowmass2021 Whitepaper: Muonium to antimuonium conversion,” in *Snowmass 2021*. 3, 2022. [arXiv:2203.11406 \[hep-ph\]](#).
 - [17] M. Artuso *et al.*, “Report of the Frontier For Rare Processes and Precision Measurements,” [arXiv:2210.04765 \[hep-ex\]](#).
 - [18] S. Bansal, G. Paz, A. Petrov, M. Tammara, and J. Zupan, “Enhanced neutrino polarizability,” *JHEP* **05** (2023) 142, [arXiv:2210.05706 \[hep-ph\]](#).
 - [19] C. M. Grant, A. Gunawardana, and A. A. Petrov, “Semileptonic decays of heavy mesons with artificial neural networks,” *Phys. Rev. D* **102** no. 3, (2020) 034003, [arXiv:1912.09058 \[hep-ph\]](#).
 - [20] **Majorana Collaboration** Collaboration, I. J. Arnuist, F. T. Avignone, A. S. Barabash, C. J. Barton, P. J. Barton, K. H. Bhimani, E. Blalock, B. Bos, M. Busch, M. Buuck, T. S. Caldwell, Y.-D. Chan, C. D. Christofferson, P.-H. Chu, M. L. Clark, C. Cuesta, J. A. Detwiler, Y. Efremenko, H. Ejiri, S. R. Elliott, G. K. Giovanetti, M. P. Green, J. Gruszko, I. S. Guinn, V. E. Guiseppe, C. R. Haufe, R. Henning, D. Hervas Aguilar, E. W. Hoppe, A. Hostiuc, M. F. Kidd, I. Kim, R. T. Kouzes, T. E. Lannen V., A. Li, A. M. Lopez, J. M. López-Castaño, E. L. Martin, R. D. Martin, R. Massarczyk, S. J. Meijer, S. Mertens, T. K. Oli, G. Othman, L. S. Paudel, W. Pettus, A. W. P.

- Poon, D. C. Radford, A. L. Reine, K. Rielage, N. W. Ruof, D. C. Schaper, D. Tedeschi, R. L. Varner, S. Vasilyev, J. F. Wilkerson, C. Wiseman, W. Xu, C.-H. Yu, and B. X. Zhu, “Final result of the majorana demonstrator’s search for neutrinoless double- β decay in ^{76}Ge ,” *Phys. Rev. Lett.* **130** (Feb, 2023) 062501. <https://link.aps.org/doi/10.1103/PhysRevLett.130.062501>.
- [21] **Majorana Collaboration** Collaboration, I. J. Arnquist, F. T. Avignone, A. S. Barabash, C. J. Barton, K. H. Bhimani, E. Blalock, B. Bos, M. Busch, M. Buuck, T. S. Caldwell, Y.-D. Chan, C. D. Christofferson, P.-H. Chu, M. L. Clark, C. Cuesta, J. A. Detwiler, Y. Efremenko, H. Ejiri, S. R. Elliott, G. K. Giovanetti, M. P. Green, J. Gruszko, I. S. Guinn, V. E. Guiseppe, C. R. Haufe, R. Henning, D. Hervas Aguilar, E. W. Hoppe, A. Hostiuc, M. F. Kidd, I. Kim, R. T. Kouzes, T. E. Lannen V., A. Li, A. M. Lopez, J. M. López-Castaño, E. L. Martin, R. D. Martin, R. Massarczyk, S. J. Meijer, T. K. Oli, G. Othman, L. S. Paudel, W. Pettus, A. W. P. Poon, D. C. Radford, A. L. Reine, K. Rielage, N. W. Ruof, D. C. Schaper, D. Tedeschi, R. L. Varner, S. Vasilyev, J. F. Wilkerson, C. Wiseman, W. Xu, C.-H. Yu, and B. X. Zhu, “Search for solar axions via axion-photon coupling with the majorana demonstrator,” *Phys. Rev. Lett.* **129** (Aug, 2022) 081803. <https://link.aps.org/doi/10.1103/PhysRevLett.129.081803>.
- [22] **Majorana Collaboration** Collaboration, I. J. Arnquist, F. T. Avignone, A. S. Barabash, C. J. Barton, K. H. Bhimani, E. Blalock, B. Bos, M. Busch, M. Buuck, T. S. Caldwell, Y.-D. Chan, C. D. Christofferson, P.-H. Chu, M. L. Clark, C. Cuesta, J. A. Detwiler, Y. Efremenko, H. Ejiri, S. R. Elliott, G. K. Giovanetti, M. P. Green, J. Gruszko, I. S. Guinn, V. E. Guiseppe, C. R. Haufe, R. Henning, D. Hervas Aguilar, E. W. Hoppe, A. Hostiuc, M. F. Kidd, I. Kim, R. T. Kouzes, T. E. Lannen V., A. Li, A. M. Lopez, J. M. López-Castaño, E. L. Martin, R. D. Martin, R. Massarczyk, S. J. Meijer, S. Mertens, T. K. Oli, G. Othman, L. S. Paudel, W. Pettus, A. W. P. Poon, D. C. Radford, J. Rager, A. L. Reine, K. Rielage, N. W. Ruof, D. C. Schaper, D. Tedeschi, R. L. Varner, S. Vasilyev, J. F. Wilkerson, C. Wiseman, W. Xu, C.-H. Yu, and B. X. Zhu, “Exotic dark matter search with the majorana demonstrator,” *Phys. Rev. Lett.* **132** (Jan, 2024) 041001. <https://link.aps.org/doi/10.1103/PhysRevLett.132.041001>.
- [23] D. Carman, S. A. Dytman, R. Magahiz, M. D. Mestayer, R. Miskiman, J. A. Mueller, R. A. Schumacher, D. J. Tedeschi, and R. A. Thompson, “The region one drift chamber for the clas spectrometer,” *Nucl. Inst. and Meth. A* **419** (1998) 315.
- [24] M. Mestayer, D. S. Carman, B. Asavapibhop, F. J. Barbosa, P. Bonneau, S. B. Christo, G. E. Dodge, T. Dooling, W. S. Duncan, S. A. Dytman, R. Feuerbach, G. P. Gilfoyle, V. Gyurjyan, K. H. Hicks, R. S. Hicks, C. E. Hyde-Wright, G. Jacobs, A. Klein, F. J. Klein, M. V. Kossov, S. E. Kuhn, R. Magahiz, R. W. Major, C. Martin, T. McGuckin, J. McNabb, R. A. Miskiman, J. A. Mueller, B. B. Niczyporuk, J. Omeara, L. M. Qin, B. A. Raue, J. Robb, F. Roudot, R. A. Schumacher, D. J. Tedeschi, R. A. Thompson, D. Tilles, W. Tuzel, K. VanSyoc, M. F. Vinyard, L. B. Weinstein, G. R. Wilkin, A. Yegneswaran, and J. Yun, “The clas drift chamber system,” *Nucl. Inst. and Meth. A* **449** (2000) 81.
- [25] A. Dzyubak, C. Djalali, N. Recalde, and D. Tedeschi, “Design of internal superconducting holding magnet for the jlab hall-b frozen spin polarized target,” *Nucl. Inst. and Meth. A* **526** (2004) 132.
- [26] B. Wojtsekhowski, D. Tedeschi, and B. Vlahovic, “A pair polarimeter for linearly polarized high energy photons,” *Nucl. Inst. and Meth. A* **515** (2003) 605.
- [27] C. de Jager, B. Wojtsekhowski, D. Tedeschi, B. Vlahovic, D. Abbott, J. Asai, G. Feldman, T. Hotta, M. Khandaker, H. Kohri, T. Matsumara, T. Mibe, T. Nakano, V. Nelyubin, G. Orielly, A. Rudge, P. Weilhammer, M. Wood, T. Yorita, and R. Zegers, “A pair polarimeter for linearly polarized high energy photons,” *Eur. Phys. Jour. A* **S19** (2004) 275.
- [28] B. McKinnon and The CLAS Collaboration[?], “Search for the Θ^+ Pentaquark in the reaction $\gamma d \rightarrow K^- K^+ n$,” *Phys. Rev. Lett.* **96** (2006) 212001.
- [29] T. Mibe and The CLAS Collaboration[?], “Coherent phi meson photoproduction from the deuteron at low energies,” *Phys. Rev. C* **76** (2007) 052202.
- [30] X. Qian and The CLAS Collaboration[?], “The extraction of the phi-n total cross section from $d(\gamma, p K^+ K^-) n$,” *Phys. Lett. B* **680** (2009) 417.
- [31] X. Qian and The CLAS Collaboration[?], “Near-threshold photoproduction of phi mesons from deuterium,” *Phys. Lett. B* **696** (2011) 338.

- [32] **CLAS Collaboration** Collaboration, M. H. Wood, R. Nasseripour, M. Paolone, C. Djalali, D. P. Weygand, K. P. Adhikari, M. Anghinolfi, J. Ball, M. Battaglieri, V. Batourine, I. Bedlinskiy, M. Bellis, B. L. Berman, A. S. Biselli, D. Branford, W. J. Briscoe, W. K. Brooks, V. D. Burkert, S. L. Careccia, D. S. Carman, P. L. Cole, P. Collins, V. Crede, A. D’Angelo, A. Daniel, R. De Vita, E. De Sanctis, A. Deur, B. Dey, S. Dhamija, R. Dickson, D. Doughty, R. Dupre, H. Egiyan, A. El Alaoui, L. El Fassi, P. Eugenio, S. Fegan, M. Y. Gabrielyan, M. Garçon, G. P. Gilfoyle, K. L. Giovanetti, F. X. Girod, J. T. Goetz, W. Gohn, R. W. Gothe, L. Graham, M. Guidal, L. Guo, K. Hafidi, H. Hakobyan, C. Hanretty, N. Hassall, K. Hicks, M. Holtrop, Y. Ilieva, D. G. Ireland, B. S. Ishkhanov, S. S. Jawalkar, H. S. Jo, K. Joo, D. Keller, M. Khandaker, P. Khetarpal, A. Kim, W. Kim, A. Klein, F. J. Klein, P. Konczykowski, V. Kubarovskiy, S. V. Kuleshov, V. Kuznetsov, K. Livingston, D. Martinez, M. Mayer, J. McAndrew, M. E. McCracken, B. McKinnon, C. A. Meyer, T. Mineeva, M. Mirazita, V. Mokeev, B. Moreno, K. Moriya, B. Morrison, E. Munevar, P. Nadel-Turonski, A. Ni, S. Niccolai, G. Niculescu, I. Niculescu, M. R. Niroula, M. Osipenko, A. I. Ostrovidov, R. Paremuzyan, K. Park, S. Park, E. Pasyuk, S. Anefalos Pereira, S. Pisano, O. Pogorelko, S. Pozdniakov, J. W. Price, S. Procureur, Y. Prok, D. Protopopescu, B. A. Raue, G. Ricco, M. Ripani, G. Rosner, P. Rossi, F. Sabatié, M. S. Saini, J. Salamanca, C. Salgado, D. Schott, R. A. Schumacher, E. Seder, H. Seraydaryan, Y. G. Sharabian, G. D. Smith, D. I. Sober, D. Sokhan, S. Stepanyan, S. S. Stepanyan, P. Stoler, I. I. Strakovsky, S. Strauch, M. Taiuti, W. Tang, C. E. Taylor, D. J. Tedeschi, S. Tkachenko, M. Ungaro, B. Vernarsky, M. F. Vineyard, E. Voutier, D. P. Watts, L. B. Weinstein, J. Zhang, B. Zhao, and Z. W. Zhao, “Absorption of the ω and ϕ mesons in nuclei,” *Phys. Rev. Lett.* **105** (Sep, 2010) 112301. <https://link.aps.org/doi/10.1103/PhysRevLett.105.112301>.
- [33] “Homepage of the snowmass21 study.” <https://snowmass21.org/rare/start>, 2021. [Online; accessed 1-October-2021].
- [34] “Homepage of the quarkonium working group.” <http://www.qwg.to.infn.it>, 2015. [Online; accessed 1-October-2021].
- [35] H. W. Fearing and M. S. Welsh, “Radiative muon capture in medium heavy nuclei in a relativistic mean field theory model,” *Phys. Rev. C* **46** (1992) 2077–2089.
- [36] M. Lee and M. MacKenzie, “Muon to Positron Conversion,” *Universe* **8** no. 4, (2022) 227, [arXiv:2110.07093 \[hep-ex\]](#).
- [37] R. Plestid and R. J. Hill, “High energy spectrum of internal positrons from radiative muon capture on nuclei,” *Phys. Rev. D* **103** no. 3, (2021) 033002, [arXiv:2010.09509 \[hep-ph\]](#).
- [38] W. C. Haxton, E. Rule, K. McElvain, and M. J. Ramsey-Musolf, “Nuclear-level effective theory of $\mu \rightarrow e$ conversion: Formalism and applications,” *Phys. Rev. C* **107** no. 3, (2023) 035504, [arXiv:2208.07945 \[nucl-th\]](#).
- [39] W. C. Haxton and E. Rule, “Distinguishing Charged Lepton Flavor Violation Scenarios with Inelastic $\mu \rightarrow e$ Conversion,” *Phys. Rev. Lett.* **133** no. 26, (2024) 261801, [arXiv:2404.17166 \[hep-ph\]](#).
- [40] D. Fontes and R. Szafron, “An effective field theory for muon conversion and muon decay-in-orbit,” [arXiv:2412.05702 \[hep-ph\]](#).
- [41] V. Cirigliano, R. Kitano, Y. Okada, and P. Tuzon, “On the model discriminating power of $\mu \rightarrow e$ conversion in nuclei,” *Phys. Rev. D* **80** (2009) 013002, [arXiv:0904.0957 \[hep-ph\]](#).
- [42] B. Bhattacharya, S. Carey, E. O. Cohen, and G. Paz, “Quasielastic Lepton-Nucleus Scattering and the Correlated Fermi Gas Model,” [arXiv:2405.05342 \[hep-ph\]](#).
- [43] M. Heinz, M. Hoferichter, T. Miyagi, F. Nöel, and A. Schwenk, “Ab initio calculations of overlap integrals for $\mu \rightarrow e$ conversion in nuclei,” [arXiv:2412.04545 \[nucl-th\]](#).
- [44] D. E. Kharzeev, “Mass radius of the proton,” *Phys. Rev. D* **104** no. 5, (2021) 054015, [arXiv:2102.00110 \[hep-ph\]](#).
- [45] S. Davidson, Y. Kuno, Y. Uesaka, and M. Yamanaka, “Probing $\mu e \gamma \gamma$ contact interactions with $\mu \rightarrow e$ conversion,” *Phys. Rev. D* **102** no. 11, (2020) 115043, [arXiv:2007.09612 \[hep-ph\]](#).
- [46] M. Pospelov and T. ter Veldhuis, “Direct and indirect limits on the electromagnetic form-factors of WIMPs,” *Phys. Lett. B* **480** (2000) 181–186, [arXiv:hep-ph/0003010](#).

- [47] N. Weiner and I. Yavin, “How Dark Are Majorana WIMPs? Signals from MiDM and Rayleigh Dark Matter,” *Phys. Rev. D* **86** (2012) 075021, [arXiv:1206.2910 \[hep-ph\]](#).
- [48] G. Paz, A. A. Petrov, M. Tammaro, and J. Zupan, “Shining dark matter in Xenon1T,” *Phys. Rev. D* **103** no. 5, (2021) L051703, [arXiv:2006.12462 \[hep-ph\]](#).
- [49] G. Ovanessian and L. Vecchi, “Direct detection of dark matter polarizability,” *JHEP* **07** (2015) 128, [arXiv:1410.0601 \[hep-ph\]](#).
- [50] E. Rule, W. C. Haxton, K. McElvain, and K. McElvain, “Nuclear-Level Effective Theory of $\mu \rightarrow e$ Conversion,” *Phys. Rev. Lett.* **130** no. 13, (2023) 131901, [arXiv:2109.13503 \[hep-ph\]](#).
- [51] V. Cirigliano, W. Dekens, J. de Vries, M. L. Graesser, and E. Mereghetti, “A neutrinoless double beta decay master formula from effective field theory,” *JHEP* **12** (2018) 097, [arXiv:1806.02780 \[hep-ph\]](#).
- [52] J. M. Arnold, B. Fornal, and M. B. Wise, “Simplified models with baryon number violation but no proton decay,” *Phys. Rev. D* **87** (2013) 075004, [arXiv:1212.4556 \[hep-ph\]](#).
- [53] A. Helset, C. Murgui, and M. B. Wise, “Simple models with both baryon and lepton number violation by two units,” *Phys. Rev. D* **104** no. 1, (2021) 015029, [arXiv:2104.03316 \[hep-ph\]](#).
- [54] S. Forte, L. Garrido, J. I. Latorre, and A. Piccione, “Neural network parametrization of deep inelastic structure functions,” *JHEP* **05** (2002) 062, [arXiv:hep-ph/0204232](#).
- [55] R. H. Bernstein, “The mu2e experiment,” *Frontiers in Physics* **7** (Jan., 2019) .
<http://dx.doi.org/10.3389/fphy.2019.00001>.